

CHIRAG RAJPUROHIT

Linux Device Driver & Firmware Engineer | Kernel Debugging | RTOS
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PROFESSIONAL SUMMARY

Linux Device Driver and Firmware Engineer with 1.5+ years of hands-on experience designing, developing, and debugging kernel-space and user-space drivers for embedded platforms. Deep expertise in Linux kernel internals — memory management, scheduler, interrupt subsystem, and device tree-based driver binding. Proven ability to debug complex hardware/software boundary issues using GDB, KGDB, strace, perf, and kernel oops/panic analysis. Experienced in firmware bring-up on ARM-based platforms (STM32, ESP32, Raspberry Pi, BeagleBone Black), platform initialization flows, and RTOS firmware development using FreeRTOS. Skilled in Python and shell automation for validation workflows, regression tooling, and production deployment pipelines. Passionate about platform reliability, systematic debugging, and continuous improvement of development practices — directly aligned with Synopsys HAV's mission.

CORE TECHNICAL SKILLS

Languages: C (primary), C++, Python, Perl, Shell/Bash Scripting

Linux Driver Development: Character Drivers, Platform Drivers, I2C, SPI, GPIO, UART drivers; VFS file operations (open, read, write, ioctl, release); kernel/user-space interfaces via /dev nodes and ioctl

Kernel Internals: Memory Management (kmalloc, kfree, copy_to/from_user), Scheduler, Interrupt Subsystem, PREEMPT_RT, Device Model (bus–driver binding, probe/remove lifecycle)

Device Tree: Device tree overlays for driver binding, hardware description, bus–driver initialization on ARM platforms

Platform Bring-Up: Hardware bring-up on Raspberry Pi, BeagleBone Black, STM32 (F103/F407), ESP32; board initialization, sensor integration, boot-time service configuration

Firmware Development: FreeRTOS firmware: task scheduling, semaphores, mutexes, message queues, IPC, interrupt management, OTA firmware update pipelines

Debugging & Validation: GDB, KGDB, strace, ltrace, perf, coredump analysis, kernel oops/panic analysis, PREEMPT_RT latency profiling, test harnesses for driver validation

Automation & Tooling: Python automation pipelines, shell scripting, systemd service orchestration, udev rules, Makefile/Kbuild, Yocto (basic), Git

Communication Protocols: I2C, SPI, UART, GPIO, TCP/IP sockets, MQTT, Wi-Fi Direct

Cloud & IoT: AWS IoT Core, MQTT, OTA Updates, X.509 Certificate Provisioning (ESP32 NVS)

PROFESSIONAL EXPERIENCE

Design Engineer – Linux Device Drivers & Firmware | Pronesis Technologies Pvt Ltd, Ahmedabad

Jan

2025 – Present

- **Driver Design & Ownership:** Design, develop, and own Linux kernel device drivers for platform hardware including GPIO, I2C, SPI, and UART; implement full kernel/user-space interfaces via VFS file operations and ioctl; manage /dev node creation through udev rules.
- **Platform Bring-Up:** Perform hardware bring-up and board initialization on ARM-based embedded platforms; integrate sensors and peripherals via device tree overlays and Linux Device Model (bus–driver binding, probe/remove/suspend/resume lifecycle).

- **Firmware Development:** Develop FreeRTOS firmware on STM32 and ESP32 supporting board initialization, peripheral configuration, and operational readiness; implement IPC, semaphores, mutexes, queues, and power management modes.
- **Debugging Across Hardware/Software Boundaries:** Debug complex issues spanning kernel, driver, and hardware layers using KGDB, coredump analysis, strace, perf, and kernel log tracing; systematically isolate driver initialization failures, memory faults, and interrupt anomalies.
- **Validation & Regression Workflows:** Build and maintain driver validation test harnesses and regression scripts; validate correctness across kernel/user-space data paths using GDB and strace; ensure stability and performance across platform releases.
- **Automation & Tooling:** Develop Python and shell automation scripts for deployment, validation, and regression workflows; author systemd service units and udev automation for reproducible platform initialization.
- **Secure & Quality Code:** Write production-grade C code following embedded coding standards; perform static analysis and continuously improve debug methodologies and code quality practices.

Embedded Engineer Intern | Pronesis Technologies Pvt Ltd, Ahmedabad

Jul 2024 – Dec 2024

- Implemented Linux character device drivers supporting multiple concurrent device instances; developed complete VFS file operations (open, read, write, ioctl, release) with copy_to_user/copy_from_user and kmalloc/kfree kernel memory management.
- Applied Linux Device Model concepts: bus–driver binding, probe/remove lifecycle, and driver registration within kernel subsystem hierarchy; created device tree entries for hardware description and driver binding.
- Validated driver correctness using strace, ltrace, and kernel log analysis; traced system calls and library interactions to isolate driver-application interface bugs — building systematic debugging discipline.

Embedded System Engineer | Episodics Labs Pvt Ltd, Ahmedabad

Jan 2024 – Jun 2024

- Performed platform bring-up and hardware integration of Panasonic PAPIR sensor and camera module with Raspberry Pi via I2C, TCP/IP socket, and Wi-Fi Direct; debugged intermittent hardware connectivity issues using GDB and kernel logs.
- Authored systemd service configurations and shell scripts for automated boot-time initialization and PIR interrupt-driven recording triggers; improved platform reliability and startup time through scripted initialization flows.
- Optimised multi-camera synchronization pipeline through socket-level tuning, demonstrating ability to debug and optimize complex hardware/software integrated systems.

KEY PROJECTS

Linux BMP280 Sensor Driver – I2C & SPI (Kernel Space)

- Designed and implemented a Linux platform device driver for BMP280 with dual-bus support (I2C and SPI); implemented register access, sensor initialization, and data acquisition using kernel bus subsystem APIs.
- Integrated via device tree overlays and Linux Device Model; demonstrated full driver lifecycle (probe, remove, suspend/resume) and kernel memory management — directly representative of Synopsys HAV driver development scope.

Virtual Character Device Driver – Multi-Device, Multi-Instance

- Developed a Linux virtual character device driver for multiple concurrent device instances; implemented VFS file operations (open, read, write, ioctl, release) with copy_to_user/copy_from_user and kmalloc/kfree memory handling.

- Validated kernel–user space data path correctness using GDB and strace; demonstrated deep understanding of kernel memory regions, /dev node interaction, and driver initialization flows.

ESP32 Patient Monitoring Device – FreeRTOS Firmware + AWS IoT

- Designed multi-task FreeRTOS firmware integrating TCAM-Mini thermal camera and mmWave sensor; applied semaphores, mutexes, inter-task queues, and event groups for safe concurrent sensor data processing.
- Implemented OTA firmware update pipeline over AWS IoT Core with secure X.509 certificate management — demonstrating firmware update/verification processes and version alignment practices.

Automated ESP32 Secure Provisioning Tool (Python Automation)

- Built a Python production automation tool to flash firmware, extract MAC IDs, register devices with AWS IoT, and provision X.509 certificates via UART — a full automated bring-up and deployment workflow.
- Demonstrates scripting-driven automation of firmware install/upgrade flows and release robustness — directly aligned with Synopsys HAV's automation and deployment requirements.

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering | 2024

Vishwakarma Government Engineering College | CGPA: 7.64 / 10

CERTIFICATIONS & TRAINING

- Mastering RTOS: Hands-on FreeRTOS and STM32Fx with Debugging – Udemy
- Linux Kernel Internals and Linux Device Driver Development
- Device Driver Debugging Techniques
- Embedded C Programming – Pantech University